

# AN ENERGY BOUND FOR AVOIDANCE INDICATORS, AND FOURIER TAILS OF DNFs OVER PRODUCT SPACES

ANONYMOUS

ABSTRACT. Let  $F$  be the indicator that none of a finite family of cylinder events occurs, on an arbitrary finite product probability space, and let  $F = \sum_U F^{=U}$  be its Efron–Stein decomposition. We prove that

$$\sum_U \left( \prod_{v \in U} \lambda_v \right) \|F^{=U}\|_2^2 \leq 1$$

for all weights  $\lambda_v \geq 1$  such that  $\prod_{v \in \text{supp } E} \lambda_v \leq 2$  for every event  $E$ . There is no dependence on the number of events, the alphabet sizes or the measure. In particular every width- $k$  DNF  $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$  satisfies  $\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k}$ , under every product measure; so  $g$  is  $\varepsilon$ -concentrated up to degree  $k \log_2(4/\varepsilon)$ . The proof does not use a switching lemma. It bounds the Efron–Stein energies by signed cover counts of the events that hold at a random point, writes the resulting hypergraph functional as the second moment of an alternating union sum under a random deletion, and bounds that sum by deletion–contraction. Over general product spaces the base  $2^{-1/k}$  is sharp, and the threshold 2 is necessary. We also give a version with weights that grow along a filtration of each coordinate, and we explain the application that motivated it: the junta term in a lower bound for the least “multiplier modulus” of the Erdős–Straus equation.

## 1. INTRODUCTION

Let  $\Omega = \prod_{v \in V_0} \Omega_v$  be a finite product of finite sets with a product probability measure  $\pi = \bigotimes_v \pi_v$ . A *cylinder event* fixes the values of the coordinates in a set  $\text{supp } E \subseteq V_0$ , its *support*. For a finite family  $\mathcal{E}$  of cylinder events let

$$F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$$

be its *avoidance indicator*, and let  $F = \sum_{U \subseteq V_0} F^{=U}$  be the Efron–Stein (Hoeffding) decomposition of  $F$  in  $L^2(\pi)$  [7, 12], [16, §8.3]. For the uniform measure on  $\{\pm 1\}^n$  this is the Fourier expansion,  $F^{=U} = \hat{F}(U) \chi_U$ . We study the weighted energy

$$G_F(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|F^{=U}\|^2, \quad \lambda^U = \prod_{v \in U} \lambda_v, \quad \lambda_v \geq 1.$$

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For constant weights  $\lambda_v = \rho^2$  it is  $\|T_\rho F\|^2$ , where  $T_\rho$  multiplies the component  $F^{=U}$  by  $\rho^{|U|}$ ; so we are asking for the largest noise operator *with*  $\rho > 1$  that still maps every avoidance indicator (of a suitable family) into the unit ball of  $L^2$ . (It is not a contraction in  $L^2$ :  $\|T_\rho F\| > \|F\|$  for nonconstant  $F$ .)

**Theorem 1.1** (Energy bound). *Let  $\lambda_v \geq 1$  ( $v \in V_0$ ), and suppose that  $\lambda^{\text{supp } E} \leq 2$  for every  $E \in \mathcal{E}$ . Then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^{=U}\|^2 \leq 1.$$

The hypothesis is on single events only. There is no condition on the number of events, on their masses or overlaps, on the alphabet sizes  $|\Omega_v|$  or on the measure. The constant 2 cannot be raised (Proposition 2.3). Since  $\sum_{U: \lambda^U > e^\tau} \|F^{=U}\|^2 \leq e^{-\tau} G_F(\lambda)$ , the theorem is a family of tail bounds, one for each choice of weights. Constant weights give the following.

**Corollary 1.2** (Energy tail). *If every event of  $\mathcal{E}$  has  $|\text{supp } E| \leq k$  ( $k \geq 1$ ), then for  $f \in \{F, 1 - F\}$  and every integer  $t \geq 0$ ,*

$$\text{En}(f; t) := \sum_{|U| > t} \|f^{=U}\|^2 \leq 2^{-(t+1)/k} \mathbb{P}(F = 0)(2 - \mathbb{P}(F = 0)) \leq 2^{-(t+1)/k}.$$

A width- $k$  DNF is  $g = 2F - 1$  for the family of its terms (with the convention  $-1 = \text{True}$ ), and the corollary becomes a statement about its Fourier tail.

**Corollary 1.3** (Fourier tails of DNFs). *Let  $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$  be computed by a DNF of width  $k$ , and let  $\pi$  be any product measure on  $\{\pm 1\}^n$  (Efron–Stein components in  $L^2(\pi)$ ; Fourier coefficients if  $\pi$  is uniform). Then for every integer  $t \geq 0$*

$$W^{>t}[g] := \sum_{|U| > t} \|g^{=U}\|^2 \leq 4p(2-p) \cdot 2^{-(t+1)/k} \leq 4 \cdot 2^{-(t+1)/k},$$

where  $p = \mathbb{P}[g = -1]$ . So  $g$  is  $\varepsilon$ -concentrated up to degree  $k \log_2(4/\varepsilon)$ . Moreover  $\sum_U 2^{|U|/k} \|g^{=U}\|^2 \leq 1 + 4p$ , and the total influence  $\mathbf{I}[g] := \sum_U |U| \|g^{=U}\|^2$  is at most  $(4k/\ln 2)p$ . The same holds for  $q$ -ary DNFs, with literals  $x_v \in S_v$ , under any product measure on  $\prod_v [q_v]$ .

*Comparison.* The classical bound goes back to Linial, Mansour and Nisan [14], via Håstad’s switching lemma [9]: width- $w$  DNFs are  $\varepsilon$ -concentrated up to degree  $O(w \log(1/\varepsilon))$ . In the form of O’Donnell’s book [16, §4.4] it reads  $W^{\geq t}[g] \leq 2 \cdot 2^{-t/(20w)}$  (we quote the constant 20 from memory and could not re-check it). It has been used in this form, with an unspecified constant  $C$ , in later work on DNFs, for instance by Mansour [15] (from memory) and by Lecomte and Tan [13, Fact 6]. Corollary 1.3 has the constant 1 in the exponent and holds for arbitrary product measures and alphabets. Over general product spaces the base  $2^{-1/k}$  is sharp: no bound  $C\rho^{-t/k}$  with  $\rho > 2$  holds, and at most a factor of order  $\sqrt{t/k}$  can be gained (Proposition 6.1).

On the uniform cube we only know that the true rate lies between  $3^{-t/k}$  and  $2^{-t/k}$  (Example 6.2). The influence bound is *not* a gain: an elementary certificate argument gives  $\mathbf{I}[g] \leq 2kp$  on the uniform cube and  $\sum_v \|L_v g\|^2 \leq 4kp$  under any product measure (Remark 5.1), so our  $(4k/\ln 2)p$  is weaker by the factor  $2/\ln 2$ , resp.  $1/\ln 2$ . Sharper bounds without the factor  $p$  are known (from memory:  $\mathbf{I} \leq 2w$  is usually attributed to Boppana [5], and we recall that Amano [1] improved it to  $\mathbf{I} \leq w$ ). We state the influence bound only because it falls out.

The nearest prior art we know is the paper of Lecomte and Tan [13]. They bound  $|\hat{g}(S)|$  by  $2^{|S|}$  times the probability that  $S$  is covered by the terms satisfied at a random point. This is the device of our Lemma 3.1, but there are three differences: our cover counts are *signed*, and vanish as soon as a satisfied term misses  $S$ ; we work in  $L^2$  with all levels and weights at once; and our degree concentration does not go through the switching lemma. The two combinatorial steps of the proof, a polarization identity (Lemma 4.1) and a deletion-contraction bound (Lemma 4.2), have no counterpart there.

*Why the switching-lemma constant is large (heuristic).* The following re-derivation uses two inputs quoted from memory: Håstad's bound  $\mathbb{P}_\rho[\text{DT}(g|_\rho) \geq d] \leq (Apw)^d$  with  $A \approx 5$  for a  $p$ -random restriction [9], [16, §4.4], and the LMN reduction  $W^{\geq t}[g] \leq 2\mathbb{P}_\rho[\text{DT}(g|_\rho) \geq pt/2]$  [14]. With  $d = pt/2$  and  $x = Apw$  the bound is  $2 \exp(\frac{t}{2Aw} x \ln x)$ , which is smallest at  $x = 1/e$ , giving  $W^{\geq t}[g] \leq 2 \cdot 2^{-t/(c_0 w)}$  with  $c_0 = 2eA \ln 2 \approx 18.8$ . This is consistent with the constant 20 quoted above. The loss is intrinsic to the route: the restriction keeps only a  $p$ -fraction of the degree, and the switching base forces  $pw \lesssim 1/(eA)$ .

*What we checked.* We have no internet access while writing this draft, and our literature search is partial. Earlier, [13] and Tal [20] were checked online; in [20] we found no explicit-constant tail bound for width- $w$  DNFs. Everything else in this section is quoted from memory and is marked as such. We believe Theorem 1.1 and the constant 1 in Corollary 1.3 are new, but we make no priority claim. Before submission the following must be checked; all descriptions in this list are from memory and may be inaccurate.

- Sharpenings of LMN and of the switching lemma: Håstad's "slight sharpening of LMN" [10]; Razborov's combinatorial proof and Beame's exposition [3]; Rossman's entropy proof [18]; Håstad's work on the correlation of parity with small-depth circuits [11]. As far as we recall these improve the dependence on size and depth for  $\text{AC}^0$ , and none states a width- $w$  DNF tail with exponent constant 1.
- Product distributions: Furst, Jackson and Smith [8] extended LMN-type concentration to product distributions on  $\{\pm 1\}^n$ , with a dependence on the bias that we do not remember; Blais, O'Donnell and Wimmer [4] study concentration and learning over arbitrary finite product domains. Corollary 1.3 has no bias dependence at all. A comparison with these papers is needed.

- Total influence of  $k$ -CNFs and DNFs: Boppana [5], Amano [1] (see Remark 5.1).
- Fourier growth and noise operators with  $\rho > 1$ :  $G_F$  at constant weight is  $\|T_\rho F\|^2$  with  $\rho > 1$ . Bounds of Fourier-growth type for  $\text{AC}^0$  are in [20] and, we believe, in later work of Chattopadhyay, Hatami, Hosseini and Lovett on pseudorandom generators from Fourier growth [6]. We have not checked whether an inequality like Theorem 1.1 appears in this literature.

*Proof outline.* Write  $G_F = \sum_V \mu^V \|L_V F\|^2$  with  $\mu_v = \lambda_v - 1$  and  $L_V = \prod_{v \in V} (I - \mathbb{E}_v)$ . Section 3 bounds  $\|L_V F\|^2$  by the mean square of a signed count  $N_{\mathcal{H}(x)}(V)$  of covers of  $V$  by the events that hold at  $x$ , and so reduces Theorem 1.1 to the hypergraph inequality  $Q_\mu(\mathcal{H}) = \sum_V \mu^V N_{\mathcal{H}}(V)^2 \leq 1$ . Section 4 proves it:  $Q_\mu(\mathcal{H})$  is the second moment of the alternating sum  $\Theta_\lambda(\mathcal{H}_P) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}}$  over the edges that miss a random set  $P$ , and deletion-contraction gives  $|\Theta| \leq 1$ . Nothing in the proof is specific to two-point spaces or to the uniform measure.

*Organisation.* Section 2 sets up notation. Sections 3 and 4 prove Theorem 1.1. Section 5 proves the corollaries, Section 6 discusses sharpness, failed strengthenings and open problems, and Section 7 gives the filtration variant. Section 8 sketches the application to the Erdős–Straus equation [19]. The exact-arithmetic script `scripts/energy_note_check.py` checks the main inequalities on small cases (Section 9).

## 2. SETTING AND ELEMENTARY FACTS

*Product spaces.* Throughout,  $\Omega = \prod_{v \in V_0} \Omega_v$  is a finite product of finite sets,  $V_0$  is finite, and  $\pi = \bigotimes_v \pi_v$  is a product probability measure. We write  $\mathbb{P}$ ,  $\mathbb{E}$ ,  $\langle \cdot, \cdot \rangle$  and  $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$ . For  $W \subseteq V_0$  we write  $x_W$  for the restriction of  $x \in \Omega$  to  $W$ ,  $\Omega_W = \prod_{v \in W} \Omega_v$  and  $\pi_W = \bigotimes_{v \in W} \pi_v$ . A *cylinder event* is a set  $E = \{x : x_v = \sigma_v (v \in \text{supp } E)\}$  with  $\text{supp } E \subseteq V_0$  and  $\sigma \in \Omega_{\text{supp } E}$ . A family  $\mathcal{E}$  of cylinder events is finite, and repetitions are allowed. Its avoidance indicator is  $F = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$ , and  $h = 1 - F$  is the indicator that some event occurs. For weights  $\lambda_v \geq 1$  we put  $\mu_v = \lambda_v - 1 \geq 0$ ,  $\lambda^U = \prod_{v \in U} \lambda_v$ ,  $\mu^U = \prod_{v \in U} \mu_v$ , and  $w_E = \lambda^{\text{supp } E}$  (the *weight* of  $E$ ).

*Efron–Stein decomposition* (classical; [7, 12], [16, §8.3]). For  $\varphi : \Omega \rightarrow \mathbb{R}$  and  $W \subseteq V_0$  let  $\mathbb{E}[\varphi \mid x_W]$  be the conditional expectation given the coordinates in  $W$ , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid x_W].$$

Then  $\varphi = \sum_U \varphi^{=U}$ , the components are mutually orthogonal,  $\varphi^{=U}$  depends only on  $x_U$ , and  $\mathbb{E}_u \varphi^{=U} = 0$  for  $u \in U$ , where  $\mathbb{E}_u$  averages over the coordinate  $u$  (with respect to  $\pi_u$ ). Put  $L_u = I - \mathbb{E}_u$  and  $L_V = \prod_{u \in V} L_u$ . The  $L_u$  are commuting orthogonal projections, and  $L_V \varphi = \sum_{U \supseteq V} \varphi^{=U}$ . If  $\mathcal{D}$  is a down-closed family of subsets of  $V_0$ , then  $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$  is a linear combination

of the functions  $\mathbb{E}[\varphi \mid x_W]$ ,  $W \in \mathcal{D}$  (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{\neg U}\|^2. \quad (1)$$

For  $\mathcal{D} = \{U : |U| \leq t\}$  this is the *energy above level  $t$* ,  $\text{En}(\varphi; t)$ . It is the least value of  $\mathbb{E}(\varphi - g)^2$  over sums  $g$  of functions that each depend on at most  $t$  coordinates. For  $\pm 1$ -valued  $g$  we also write  $W^{>t}[g] = \text{En}(g; t)$ , as in [16].

**Definition 2.1** (Energy generating function). For weights  $\lambda_v \geq 1$ ,

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|\varphi^{\neg U}\|^2.$$

**Lemma 2.2** (Elementary identities). For every  $\varphi$  and all weights  $\lambda_v \geq 1$ :

- (a)  $G_\varphi(\lambda) = \sum_{V \subseteq V_0} \mu^V \|L_V \varphi\|^2$ ;
- (b)  $\sum_{U: \lambda^U > e^\tau} \|\varphi^{\neg U}\|^2 \leq e^{-\tau} G_\varphi(\lambda)$  for every real  $\tau$ ;
- (c)  $G_{1-\varphi} = 1 - 2\mathbb{E}\varphi + G_\varphi$ ; in particular  $G_F \leq 1$  if and only if  $G_h \leq 2\mathbb{E}h$ ;
- (d) if  $\varphi(x) = \prod_v \varphi_v(x_v)$  is a product, then  $G_\varphi(\lambda) = \prod_v G_{\varphi_v}(\lambda_v)$ , where  $G_{\varphi_v}(\lambda_v) = (\mathbb{E}\varphi_v)^2 + \lambda_v \text{Var} \varphi_v$ .

*Proof.* (a) Expand  $\lambda^U = \prod_{v \in U} (1 + \mu_v) = \sum_{V \subseteq U} \mu^V$  and use  $\|L_V \varphi\|^2 = \sum_{U \supseteq V} \|\varphi^{\neg U}\|^2$ . (b) is Markov's inequality. (c)  $(1 - \varphi)^{\neg U} = -\varphi^{\neg U}$  for  $U \neq \emptyset$  and  $(1 - \varphi)^{\neg \emptyset} = 1 - \mathbb{E}\varphi$ . (d) For a product,  $\varphi^{\neg U} = \prod_{v \in U} (\varphi_v - \mathbb{E}\varphi_v) \prod_{v \notin U} \mathbb{E}\varphi_v$ .  $\square$

**Proposition 2.3** (A single event; the threshold 2). Let  $E$  fix  $x_v = \sigma_v$  on  $\text{supp } E$ , put  $p_v = \pi_v(\sigma_v)$  and  $\pi_E = \mathbb{P}(E) = \prod_{v \in \text{supp } E} p_v$ . Then

$$G_{1-\mathbf{1}_E}(\lambda) = 1 - 2\pi_E + \pi_E \prod_{v \in \text{supp } E} (\lambda_v - \mu_v p_v).$$

Hence, for fixed weights with  $w_E > 2$ ,  $G_{1-\mathbf{1}_E}(\lambda) > 1$  as soon as all  $p_v > 0$  are small enough. So the constant 2 in Theorem 1.1 cannot be replaced by any larger number, even for a single event.

*Proof.* By Lemma 2.2(d), with  $\varphi_v = \mathbf{1}[x_v = \sigma_v]$ ,  $G_{\mathbf{1}_E} = \prod_v (p_v^2 + \lambda_v p_v(1 - p_v)) = \pi_E \prod_v (\lambda_v - \mu_v p_v)$ . Then use Lemma 2.2(c). The product tends to  $w_E$  as  $p \rightarrow 0$ .  $\square$

On the uniform cube  $\{\pm 1\}^n$  the single-event condition is the weaker  $\prod_v (1 + \lambda_v)/2 \leq 2$ . The threshold  $w_E \leq 2$  is forced by large alphabets or strongly biased measures.

### 3. THE COVER BOUND

A *hypergraph*  $\mathcal{H}$  is a finite set of subsets (*edges*) of  $V_0$ ; the empty edge is allowed. For  $V \subseteq V_0$  let

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}: \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \quad Q_{\mu}(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V N_{\mathcal{H}}(V)^2.$$

So  $N_{\mathcal{H}}(V)$  is a signed count of the subfamilies whose traces cover  $V$ . It depends only on the multiset of traces  $J \cap V$ . It does not change if an edge is replaced by several edges with the same trace: a class of  $m \geq 1$  edges with equal traces contributes  $\sum_{j \geq 1} \binom{m}{j} (-1)^j = -1$  to the choice of whether the class meets  $\mathcal{J}$ , exactly like a single edge. If some edge misses  $V$ , then  $N_{\mathcal{H}}(V) = 0$ : toggling that edge is a sign-reversing involution on the covering subfamilies. Also  $N_{\mathcal{H}}(\emptyset) = \mathbf{1}[\mathcal{H} = \emptyset]$ . For example,  $Q_{\mu}(\{J\}) = \lambda^J - 1$ , and for pairwise disjoint edges  $Q_{\mu}(\{J_1, \dots, J_m\}) = \prod_i (\lambda^{J_i} - 1)$ .

**Lemma 3.1** (Cover bound). *For every family  $\mathcal{E}$  of cylinder events and every  $V \subseteq V_0$ ,*

$$\|L_V F\|^2 \leq \mathbb{E}_x[N_{\mathcal{H}(x)}(V)^2], \quad \mathcal{H}(x) = \{\text{supp } E : E \in \mathcal{E}, x \in E\}.$$

*Consequently  $G_F(\lambda) \leq \mathbb{E}_x[Q_{\mu}(\mathcal{H}(x))]$  for all weights  $\lambda_v \geq 1$ .*

*Proof.* Write  $x = (\sigma, y)$  with  $\sigma \in \Omega_V$  and  $y \in \Omega_{V^c}$ ,  $V^c = V_0 \setminus V$ . The operator  $L_V$  acts on the  $V$ -coordinates only, so  $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|_{L^2(\pi_V)}^2$  with  $g_y = F(\cdot, y)$ . Now  $g_y$  is the avoidance indicator on  $\Omega_V$  of the *restricted* events: those  $E \in \mathcal{E}$  whose conditions off  $V$  hold at  $y$ , restricted to  $\text{supp } E \cap V$ . Expand

$$g_y = \prod_E (1 - \mathbf{1}_{E|_y}) = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$$

over subfamilies  $\mathcal{J}$  of restricted events, where  $\mathbf{1}_{\mathcal{J}}$  is the indicator of their intersection, which is a cylinder on  $\Omega_V$  with support  $\bigcup_{E \in \mathcal{J}} (\text{supp } E \cap V)$ , or it is empty. The projection  $L_V$  annihilates every function that does not depend on all coordinates in  $V$ , hence every cylinder whose support is a proper subset of  $V$ . A cylinder with support  $V$  is a point indicator  $\mathbf{1}_{\{\sigma\}}$ . Therefore  $L_V g_y = L_V c_y$ , where

$$c_y(\sigma) = \sum_{\mathcal{J}: \text{supp } \mathbf{1}_{\mathcal{J}} = V, \mathbf{1}_{\mathcal{J}} = \mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

Since  $L_V$  is an orthogonal projection in  $L^2(\pi_V)$ ,  $\|L_V c_y\|^2 \leq \mathbb{E}_{\sigma} c_y(\sigma)^2$ . The subfamilies counted in  $c_y(\sigma)$  are exactly the subfamilies of events that hold at  $x = (\sigma, y)$  whose traces on  $V$  cover  $V$ . Two events that hold at  $x$  and have the same support are the same cylinder. By the remark on repeated traces,  $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$ . Taking  $\mathbb{E}_y$  gives the first claim. For the second, multiply by  $\mu^V$ , sum over  $V$  and use Lemma 2.2(a).  $\square$

The lemma is an equality for  $V = \emptyset$  (both sides equal  $\mathbb{E}F$ ). The only loss is the projection step  $\|L_V c\| \leq \|c\|$ . Two features matter below. First, the suppression is built in:  $N_{\mathcal{H}(x)}(V) = 0$  as soon as some event that holds at  $x$  misses  $V$ . Second, the right-hand side no longer involves the measure, except through the law of  $\mathcal{H}(x)$ ; Theorem 1.1 will follow from a bound on  $Q_{\mu}$  that is uniform over all hypergraphs whose edges have weight at most 2.

**Remark 3.2** (Unsigned counts are too weak). Lecomte and Tan [13, Fact 9] bound  $|\hat{g}(S)|$  for a DNF  $g$  on the uniform cube by  $2^{|S|} \mathbb{P}_x[S \text{ is covered by}]$

the terms satisfied by  $x$ ]; we quote the form from [13], checked earlier. The indicator “ $S$  is covered” does not see cancellations. For example, if  $\mathcal{H}(x)$  contains a singleton  $\{u\}$  and an edge containing a set  $V \not\ni u$ , then  $N_{\mathcal{H}(x)}(V) = 0$ , while  $\mathbf{1}[V \text{ is covered}] = 1$ , so the bound of [13] charges  $2^{|V|}$  at such points.

#### 4. THE HYPERGRAPH INEQUALITY

For a finite multiset  $\mathcal{C}$  of subsets of  $V_0$  (empty members allowed) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J} \subseteq \mathcal{C}} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} \quad (\Theta_\lambda(\emptyset) = 1),$$

the sum running over sub-multisets (members are distinguishable).

**Lemma 4.1** (Polarization). *Let  $\lambda_v \geq 1$ ,  $\mu_v = \lambda_v - 1$ , and let  $P \subseteq V_0$  be random with  $\mathbb{P}(v \in P) = \mu_v/\lambda_v$ , independently for different  $v$ . For every hypergraph  $\mathcal{H}$ ,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \quad \mathcal{H}_P = \{J \in \mathcal{H} : J \cap P = \emptyset\}.$$

*Proof.* Let  $\tau(R) = \mathbf{1}[R \cap J \neq \emptyset \text{ for all } J \in \mathcal{H}]$  for  $R \subseteq V_0$ , and  $\hat{\tau}(V) = \sum_{R \subseteq V} (-1)^{|V \setminus R|} \tau(R)$ . By inclusion–exclusion,

$$\tau(R) = \sum_{\mathcal{J} \subseteq \mathcal{H}} (-1)^{|\mathcal{J}|} \mathbf{1}[R \cap \bigcup \mathcal{J} = \emptyset]. \quad (2)$$

The Möbius transform of  $R \mapsto \mathbf{1}[R \cap U = \emptyset]$  is  $V \mapsto (-1)^{|V|} \mathbf{1}[V \subseteq U]$ , so  $\hat{\tau}(V) = (-1)^{|V|} N_{\mathcal{H}}(V)$ . Let

$$\mathcal{T}(p) = \sum_V \hat{\tau}(V) p^V = \sum_R \tau(R) \prod_{v \in R} p_v \prod_{v \notin R} (1 - p_v)$$

be the multilinear extension of  $\tau$ . If the  $p_v$  are independent random variables with mean 0 and variance  $\mu_v$ , then distinct monomials are orthogonal and  $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V \hat{\tau}(V)^2 = Q_\mu(\mathcal{H})$ , whatever the laws of the  $p_v$ . Take  $p_v = 1$  with probability  $\mu_v/\lambda_v$  and  $p_v = -\mu_v$  otherwise. The mean is  $\mu_v/\lambda_v - \mu_v/\lambda_v = 0$  and the variance is  $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$ . Let  $P = \{v : p_v = 1\}$ . In the second form of  $\mathcal{T}$ , the factor  $1 - p_v$  vanishes for  $v \in P$ , so only the  $R \supseteq P$  contribute; writing  $R = P \cup B$  with  $B \subseteq P^c$ , and using  $1 - p_v = \lambda_v$  for  $v \notin P$ ,

$$\mathcal{T}(p) = \sum_{B \subseteq P^c} (-\mu)^B \lambda^{P^c \setminus B} \tau(P \cup B).$$

An edge that meets  $P$  meets  $P \cup B$ , so (2) for the hypergraph  $\mathcal{H}_P$  gives  $\tau(P \cup B) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \mathbf{1}[B \cap \bigcup \mathcal{J} = \emptyset]$ . For  $U \subseteq P^c$ ,

$$\sum_{B \subseteq P^c \setminus U} (-\mu)^B \lambda^{P^c \setminus B} = \lambda^U \prod_{v \in P^c \setminus U} (\lambda_v - \mu_v) = \lambda^U.$$

Hence  $\mathcal{T}(p) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} = \Theta_\lambda(\mathcal{H}_P)$ .  $\square$

**Lemma 4.2** (Matching bound). *If every member of  $\mathcal{C}$  has  $w_J := \lambda^J \leq 2$ , then for every matching  $\mathcal{M} \subseteq \mathcal{C}$  (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \leq \prod_{J \in \mathcal{M}} (w_J - 1); \quad \text{in particular } |\Theta_\lambda(\mathcal{C})| \leq 1.$$

*Proof.* For  $v \in V_0$  let  $\mathcal{C} - v$  be the multiset of members not containing  $v$ , and  $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$  (as a multiset). Split the sum over  $\mathcal{J}$  according to whether  $\mathcal{J}$  contains a member through  $v$ . If it does, then  $\lambda^{\bigcup \mathcal{J}} = \lambda_v \lambda^{\bigcup (\mathcal{J}/v)}$ ; the terms without such a member form  $\Theta(\mathcal{C} - v)$  both in  $\Theta(\mathcal{C})$  and in  $\Theta(\mathcal{C}/v)$ . Hence

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v [\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v \Theta(\mathcal{C}/v) - \mu_v \Theta(\mathcal{C} - v). \quad (3)$$

Both families again have all weights at most 2, since  $w_{J \setminus v} = w_J / \lambda_v$  for  $J \ni v$  and the other members are unchanged. We induct on  $|\bigcup \mathcal{C}|$ , over all finite multisets; note that  $\mathcal{C}/v$  may contain empty members (from  $J = \{v\}$ ), and these are handled by the first case below. If  $\emptyset \in \mathcal{C}$ , then  $\Theta(\mathcal{C}) = 0$  (toggling the empty member pairs the terms), and the claim holds. If  $\mathcal{C} = \emptyset$ , then  $\Theta = 1$  and  $\mathcal{M} = \emptyset$ . This covers the base case  $\bigcup \mathcal{C} = \emptyset$ . If  $\mathcal{M} = \emptyset$  and  $\mathcal{C} \neq \emptyset$ , it suffices to prove the bound for  $\mathcal{M} = \{J\}$  with any  $J \in \mathcal{C}$ , since  $w_J - 1 \leq 1$ . So let  $J_0 \in \mathcal{M}$  be nonempty,  $v \in J_0$ , and  $\Pi' = \prod_{J \in \mathcal{M} \setminus \{J_0\}} (w_J - 1)$ . In  $\mathcal{C}/v$  the family  $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$  is a matching, so by induction  $|\Theta(\mathcal{C}/v)| \leq (w_{J_0}/\lambda_v - 1)\Pi'$ . In  $\mathcal{C} - v$  the family  $\mathcal{M} \setminus \{J_0\}$  is a matching, so  $|\Theta(\mathcal{C} - v)| \leq \Pi'$ . By (3),  $|\Theta(\mathcal{C})| \leq (w_{J_0} - \lambda_v)\Pi' + \mu_v \Pi' = (w_{J_0} - 1)\Pi'$ .  $\square$

The hypothesis  $w_J \leq 2$  enters only through  $w_J - 1 \leq 1$  and its stability under contraction. Without it  $\Theta$  is unbounded: for  $m$  disjoint singletons of weight  $\lambda$ ,  $\Theta = (1 - \lambda)^m$ .

**Theorem 4.3** (Hypergraph inequality). *If every edge of a hypergraph  $\mathcal{H}$  has  $w_J = \lambda^J \leq 2$ , then*

$$Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} (w_J - 1) \quad \text{for every matching } \mathcal{M} \subseteq \mathcal{H};$$

*in particular  $Q_\mu(\mathcal{H}) \leq 1$ , and  $Q_\mu(\mathcal{H}) \leq \min_{J \in \mathcal{H}} w_J - 1$  if  $\mathcal{H} \neq \emptyset$ . Equality holds if  $\mathcal{H} = \mathcal{M}$  is itself a matching.*

*Proof.* By Lemma 4.1, and Lemma 4.2 applied to  $\mathcal{H}_P$  with the matching  $\mathcal{M} \cap \mathcal{H}_P$ ,  $Q_\mu(\mathcal{H}) \leq \mathbb{E}_P \prod_{J \in \mathcal{M} \cap \mathcal{H}_P} (w_J - 1)^2$ . The members of  $\mathcal{M}$  are disjoint, so the events  $\{J \cap P = \emptyset\}$ ,  $J \in \mathcal{M}$ , are independent, with probabilities  $\prod_{v \in J} (1 - \mu_v / \lambda_v) = 1/w_J$ . Hence

$$Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} \left[ \left(1 - \frac{1}{w_J}\right) + \frac{(w_J - 1)^2}{w_J} \right] = \prod_{J \in \mathcal{M}} (w_J - 1).$$

The equality case is the example after the definition of  $Q_\mu$ .  $\square$

*Proof of Theorem 1.1.* By Lemma 3.1,  $G_F(\lambda) \leq \mathbb{E}_x [Q_\mu(\mathcal{H}(x))]$ . Every edge of  $\mathcal{H}(x)$  is the support of an event, so it has weight at most 2, and Theorem 4.3 gives  $Q_\mu(\mathcal{H}(x)) \leq 1$  for every  $x$ .  $\square$

Theorem 4.3 with the matching bound gives slightly more:  $G_F(\lambda) \leq \mathbb{E}_x \min_{\mathcal{M}} \prod_{J \in \mathcal{M}} (w_J - 1)$ , the minimum over matchings in  $\mathcal{H}(x)$ . It would be interesting to know whether the minimum over matchings can be replaced by a fractional-matching version (Conjecture 6.5).

## 5. ENERGY TAILS AND DNFs

*Proof of Corollary 1.2.* Take  $\lambda_v = \lambda = 2^{1/k}$  for all  $v$ ; then  $w_E = \lambda^{|\text{supp } E|} \leq 2$ , and Theorem 1.1 gives  $G_F(\lambda) \leq 1$ . Let  $a = \mathbb{P}(F = 0) = \mathbb{E}h$ . Since  $\|F^{\neq \emptyset}\|^2 = (\mathbb{E}F)^2 = (1 - a)^2$ ,

$$\sum_{U \neq \emptyset} \lambda^{|U|} \|F^{\neq U}\|^2 \leq 1 - (1 - a)^2 = a(2 - a).$$

For  $t \geq 0$  every  $U$  with  $|U| > t$  is nonempty and has  $\lambda^{|U|} \geq 2^{(t+1)/k}$ , so  $\text{En}(F; t) \leq 2^{-(t+1)/k} a(2 - a)$ . Finally  $(1 - F)^{\neq U} = -F^{\neq U}$  for  $U \neq \emptyset$ .  $\square$

*Proof of Corollary 1.3.* Let  $g$  be computed by the DNF  $T_1 \vee \dots \vee T_m$  with  $-1 = \text{True}$ , and let  $E_i$  be the set of inputs that satisfy the term  $T_i$ . A term is a conjunction of at most  $k$  literals on distinct variables (a term containing  $x_v$  and  $\bar{x}_v$  is never satisfied and may be dropped), so  $E_i$  is a cylinder event with  $|\text{supp } E_i| \leq k$ . Then  $g = 2F - 1$  with  $F = \prod_i (1 - \mathbf{1}_{E_i})$ , and  $p = \mathbb{P}[g = -1] = \mathbb{P}(F = 0)$ . As  $g^{\neq U} = 2F^{\neq U}$  for  $U \neq \emptyset$ , the first claim is Corollary 1.2 multiplied by 4, and  $p(2 - p) \leq 1$ . If  $4 \cdot 2^{-(d+1)/k} \leq \varepsilon$ , i.e.  $d + 1 \geq k \log_2(4/\varepsilon)$ , then  $W^{>d}[g] \leq \varepsilon$ ; the least such integer  $d$  is at most  $k \log_2(4/\varepsilon)$ .

For the weighted sum,  $\|g^{\neq \emptyset}\|^2 = (1 - 2p)^2$  and, with  $\lambda = 2^{1/k}$ ,

$$\sum_{U \neq \emptyset} \lambda^{|U|} \|g^{\neq U}\|^2 = 4(G_h - p^2),$$

where  $G_h = G_F - 1 + 2p \leq 2p$  by Lemma 2.2(c). So the sum is at most  $(1 - 2p)^2 + 8p - 4p^2 = 1 + 4p$ . For the influence,  $\sum_U (\lambda^{|U|} - 1) \|F^{\neq U}\|^2 = G_F - \mathbb{E}F \leq p$ , and  $\lambda^{|U|} - 1 \geq |U| \ln \lambda = |U| \ln 2/k$ . Hence

$$\mathbf{I}[g] = \sum_U |U| \|g^{\neq U}\|^2 = 4 \sum_U |U| \|F^{\neq U}\|^2 \leq (4k/\ln 2) p.$$

For a  $q$ -ary DNF, a term  $\bigwedge_{v \in S} (x_v \in S_v)$  is the disjoint union of the cylinder events  $\{x_S = \sigma\}$ ,  $\sigma \in \prod_{v \in S} S_v$ , which all have support  $S$ . Replacing each term by these events changes neither  $F$  nor the supports.  $\square$

For a width- $k$  CNF  $h$ ,  $-h$  is a width- $k$  DNF and  $W^{>t}[h] = W^{>t}[-h]$ ; so the same bounds hold with  $p = \mathbb{P}[h = +1]$  (the probability that  $h$  is false), and the bound  $4 \cdot 2^{-(t+1)/k}$  holds as stated. Here  $\mathbf{I}[g] = \sum_U |U| \|g^{\neq U}\|^2 = \sum_v \|L_v g\|^2$  is the Efron–Stein total influence; it is the usual total influence for the uniform measure.

**Remark 5.1** (The influence bound is elementary). Let  $g$  be a width- $k$  DNF and  $p = \mathbb{P}[g = -1]$ . If  $g(x) = -1$ , fix a term satisfied at  $x$ ; changing a coordinate outside it leaves  $g = -1$ . So the fibre of  $x$  in direction  $v$  is

non-constant for at most  $k$  coordinates  $v$ . Since  $\|L_v g\|^2 = \mathbb{E} \text{Var}_{x_v}(g)$  and  $\text{Var}_{x_v}(g) = 4q(1-q) \leq 4q$  on a non-constant fibre, with  $q$  the conditional probability of  $g = -1$ , we get  $\sum_v \|L_v g\|^2 \leq 4kp$  under any product measure. On the uniform cube,  $\mathbf{I}[g]$  is the mean number of sensitive coordinates; every sensitive edge has exactly one endpoint with  $g = -1$ , so  $\mathbf{I}[g] \leq 2kp$ . Corollary 1.3 gives the weaker  $(4k/\ln 2)p$ .

**Remark 5.2** (Non-uniform widths). Theorem 1.1 also handles terms of very different widths, and weights adapted to the variables. For instance, if every term contains at most  $k_1$  variables from a set  $A$  and at most  $k_2$  from its complement, then  $\lambda_v = 2^{1/(2k_1)}$  on  $A$  and  $2^{1/(2k_2)}$  off  $A$  are admissible. This gives  $\sum \|F^=U\|^2 \leq 2^{-\tau}$  over the  $U$  with  $|U \cap A|/k_1 + |U \setminus A|/k_2 > 2\tau$ . The weights used in Section 8 are of this kind, with  $\log \lambda_v$  proportional to the logarithm of the size of the alphabet at  $v$ .

**Remark 5.3** (What is not claimed). Corollary 1.3 concerns concentration on low *degree* only. It says nothing about the number of significant coefficients (Mansour's sparsity bound and Mansour's conjecture [15], [13]), and it does not give the random-restriction (decision-tree depth) form of the switching lemma. It does not apply to depth-3 circuits or to  $\text{AC}^0$ ; we do not know whether the method extends to them.

## 6. SHARPNESS, LIMITS OF THE METHOD, OPEN PROBLEMS

**Proposition 6.1** (The base  $2^{-1/k}$  is sharp). *Fix  $k \geq 1$  and  $j \geq 1$ , and let  $t = jk - 1$ . For every  $\eta > 0$  there is a family of cylinder events of width  $k$ , on a product space with all alphabets of size 2 and a suitable product measure, or on  $[q]^n$  with the uniform measure and  $q$  large, such that*

$$\text{En}(F; t) \geq \frac{2^{-(t+1)/k}}{e\sqrt{j}} - \eta.$$

*Consequently, no bound  $\text{En}(F; t) \leq C\rho^{-(t+1)/k}$  with  $\rho > 2$  holds for all width- $k$  systems, and the same is true for  $W^{>t}[g]$  of width- $k$  DNFs under biased product measures on  $\{\pm 1\}^n$ .*

*Proof.* Take  $m$  events on pairwise disjoint sets  $S_1, \dots, S_m$  of  $k$  coordinates, each fixing at every coordinate of its support a value of probability  $p$  (the value  $-1$  of a  $p$ -biased bit, or one value of  $[q]$  with  $q = 1/p$ ), and put  $\pi = p^k$ . The factors  $1 - \mathbf{1}_{E_i}$  are independent, so  $F^=U = \prod_i (1 - \mathbf{1}_{E_i})^{=U \cap S_i}$ . If  $U$  is the union of  $j$  of the  $S_i$ , Lemma 2.2(d) gives  $\|F^=U\|^2 = [\pi(1-p)^k]^j (1-\pi)^{2(m-j)}$ , and  $|U| = jk$ . Hence

$$\text{En}(F; jk - 1) \geq \binom{m}{j} [\pi(1-p)^k]^j (1-\pi)^{2(m-j)} \rightarrow e^{-2s} \frac{s^j}{j!}$$

as  $p \rightarrow 0$  with  $m = \lfloor s/\pi \rfloor$ . With  $s = j/2$  and  $j! \leq e j^{j+1/2} e^{-j}$  the limit is at least  $2^{-j}/(e\sqrt{j})$ , and  $2^{-j} = 2^{-(t+1)/k}$ .  $\square$

So at most a factor of order  $\sqrt{t/k}$  could be gained in Corollary 1.2 over general product spaces. In the limit  $p \rightarrow 0$ , the supremum over  $s$  of  $e^{-2s} \sum_{i \geq j} s^i / i!$  (the limiting value of  $\text{En}(F; jk-1)$ ) behaves like  $2 \cdot 2^{-j} / \sqrt{2\pi j}$  as  $j \rightarrow \infty$  (heuristically: the term  $i = j$  at  $s \approx j/2$ , and a tail of ratio about  $1/2$ ). We do not prove this asymptotic; numerically the normalised supremum is 0.2255 at  $j = 10$  against  $2/\sqrt{20\pi} = 0.2523$ , and 0.0876 against 0.0892 at  $j = 80$  (internal computations). Together with Proposition 2.3 this shows that both the threshold 2 in Theorem 1.1 and the base of Corollary 1.2 are optimal at the level of generality of the theorem. Note that the extremal measures have atoms of probability  $p \rightarrow 0$ , depending on  $t$ : the base is sharp for the supremum over *all* product measures. For a fixed measure whose atoms are bounded below (for instance  $[q]^n$  uniform with  $q$  fixed, or the uniform cube) the proposition says nothing, and the optimal rate is open (cf. Example 6.2).

**Example 6.2** (The uniform cube). On  $\{\pm 1\}^n$  with the uniform measure the extremal families above are not available ( $p = 1/2$ ), and the truth may be better. Let  $g$  be the OR of  $s$  parities on pairwise disjoint blocks  $B_1, \dots, B_s$  of  $k$  variables. Each parity is a width- $k$  DNF (with  $2^{k-1}$  terms), so  $g$  is a width- $k$  DNF. If the  $i$ -th parity is “true” when  $\chi_{B_i} = -1$ , then  $F = \prod_i (1 + \chi_{B_i})/2$ , and  $\text{En}(F; jk-1) = 4^{-s} \sum_{i \geq j} \binom{s}{i}$ . Taking  $s$  near  $4j/3$  and using Stirling’s formula,  $\max_s 4^{-s} \binom{s}{j} = 3^{-j+o(j)}$ . So the exponential rate of the best width- $k$  bound on the uniform cube lies between  $3^{-t/k}$  and  $2^{-t/k}$ . We do not know where. In random and hill-climbing searches over small Boolean systems (earlier internal computations) we found nothing better than this example.

*No one-event-at-a-time proof.* It is natural to try to prove Theorem 1.1 by adding the events one at a time. The following example shows that this cannot work in the obvious form. Adding a further event of weight at most 2 to a system can *increase*  $G_F$ .

**Example 6.3** (Adding an event can increase  $G_F$ ). Let  $n = 2$ ,  $\Omega = [8]^2$  with the uniform measure,  $\lambda_1 = \lambda_2 = \sqrt{2}$ , and let  $\mathcal{E}$  consist of the 49 point events  $\{x = (a, b)\}$  with  $a \neq 0 \neq b$ . Let  $A = \{x = (0, 0)\}$ . All events have weight 2, and  $G_{F_{\mathcal{E}}} < G_{F_{\mathcal{E} \cup \{A\}}}$ ; numerically the values are 0.31571 and 0.31757. (The exact values are computed in Section 9.) A similar example exists on  $[5]^3$  with  $\lambda_v = 2^{1/3}$ : take the 48 point events with exactly two nonzero coordinates and add  $A = \{x = (0, 0, 0)\}$ ; then  $G_F$  increases from 0.75406 to 0.75467 (checked by the script of Section 9, with 50-digit arithmetic).

Also the inequality fails beyond the threshold. For  $n$  uniform bits, the OR of the  $n$  literals  $x_v = -1$  and  $\lambda_v = 4$  give  $G_F = (5/4)^n$  by Lemma 2.2(d) and Proposition 2.3. We state two strengthenings that are supported by numerical evidence but are open.

**Conjecture 6.4** (Excess weights). *For every family of cylinder events and all weights  $\lambda_v \geq 1$ ,*

$$G_F(\lambda) \leq \prod_{E \in \mathcal{E}} (1 + \mathbb{P}(E)(w_E - 2)_+).$$

**Conjecture 6.5** (Fractional matchings). *If every edge of  $\mathcal{H}$  has  $w_J \leq 2$ , then  $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{H}} (w_J - 1)^{y_J}$  for every fractional matching  $y$  ( $y_J \geq 0$ ,  $\sum_{J \ni v} y_J \leq 1$  for all  $v$ ).*

Both were tested numerically on small random systems (several thousand cases on  $[q]^n$  with  $q \leq 5$ ,  $n \leq 9$ ; floating-point internal computations, with a numerical linear program over  $y$  for Conjecture 6.5; not part of this note), without a violation. For Conjecture 6.4 only the weaker form  $G_F \leq \exp(\sum_E \mathbb{P}(E)(w_E - 2)_+)$  was tested. For Conjecture 6.5 equality (up to rounding) is frequent. Example 6.3 shows that a proof of Conjecture 6.4 by induction on the number of events would need a stronger induction hypothesis.

*Open problems.* (1) Determine the optimal rate for width- $k$  DNFs on the uniform cube (Example 6.2). (2) Is there an analogue of Theorem 1.1 for depth-3 circuits, or for  $\text{AC}^0$ , with a weight condition on the bottom layer? (3) Does the signed cover count give sparsity bounds in the direction of Mansour’s conjecture?

## 7. WEIGHTS ALONG A FILTRATION

In Theorem 1.1 the cost of a component  $F^{=U}$  is a product over the coordinates in  $U$ . In the application the natural cost of a function of a residue  $x_\ell \bmod \ell^f$  depends on *how many* base- $\ell$  digits it reads, and an event that fixes  $x_\ell \bmod \ell^v$  fixes the first  $v$  digits. We now prove the corresponding variant.

*Setting.* Let  $\mathcal{P}$  be a finite index set. Each coordinate  $X_\ell$  ( $\ell \in \mathcal{P}$ ) is a tuple of independent *digits*  $(\delta_{\ell,i})_{i_0(\ell) \leq i < f_\ell}$  with arbitrary distributions on finite sets, and the coordinates are independent. The Efron–Stein decomposition  $F = \sum_U F^{=U}$  is taken over the digits, so  $U$  is a set of pairs  $(\ell, i)$ ; we write  $U_\ell = \{i : (\ell, i) \in U\}$ . An event  $E$  is a *digit-prefix event* if, for some integers  $0 \leq v_\ell(E) \leq f_\ell$ , it fixes exactly the digits  $\delta_{\ell,i}$  with  $i_0(\ell) \leq i < v_\ell(E)$ , for each  $\ell$ , to arbitrary values (no digit of  $X_\ell$  if  $v_\ell(E) \leq i_0(\ell)$ ). The offsets are integers  $i_0(\ell) \geq 0$ . Such an event is a cylinder event on the digits. (The offset  $i_0(\ell)$  allows digits below  $i_0(\ell)$  to be fixed in advance; they are not coordinates.)

**Theorem 7.1** (Filtration energy bound). *Let  $\lambda_\ell \geq 1$  ( $\ell \in \mathcal{P}$ ), and suppose that every event of  $\mathcal{E}$  is a digit-prefix event with  $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$ . Then*

$$\sum_U \left( \prod_{\ell: U_\ell \neq \emptyset} \lambda_\ell^{1 + \max U_\ell} \right) \|F^{=U}\|^2 \leq 1.$$

Applied directly, Theorem 1.1 with digit weights would charge a set  $U$  for each digit in it, and the event weights would grow with the number of fixed digits. Here the charge for  $U_\ell$  depends only on its *deepest* digit, which is what a function of  $X_\ell$  that reads  $U_\ell$  costs, and the hypothesis loses only the factor 2 in the exponent.

*Proof.* Write  $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$  for  $i \geq i_0(\ell)$ ,  $\Lambda_{\ell,i_0(\ell)-1} = 1$ , and  $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$  for  $i_0(\ell) \leq j < f_\ell$ . For such  $j$  let  $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$ , where  $\mathbb{E}_{B_{\ell,j}}$  averages over the *tail block*  $B_{\ell,j} = (\delta_{\ell,i})_{i \geq j}$ . Averaging over a digit in  $U$  kills  $F^{=U}$ , and  $F^{=U}$  does not depend on the digits outside  $U$ . Hence  $P_{\ell,j}F^{=U} = F^{=U}$  if  $U_\ell \neq \emptyset$  and  $\max U_\ell \geq j$ , and  $P_{\ell,j}F^{=U} = 0$  otherwise. Call a set  $V$  of pairs  $(\ell, j)$  a *selection* if it contains at most one pair for each  $\ell$ , and put  $L'_V = \prod_{(\ell,j) \in V} P_{\ell,j}$ . Then  $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$ , where  $U \succ V$  means  $\max U_\ell \geq j$  for all  $(\ell, j) \in V$ . For fixed  $U$ , the sum of  $\mu'^V$  over the selections  $V$  with  $U \succ V$  factorises and telescopes:

$$\prod_{\ell} \left(1 + \sum_{i_0(\ell) \leq j \leq \max U_\ell} \mu'_{\ell,j}\right) = \prod_{\ell: U_\ell \neq \emptyset} \Lambda_{\ell, \max U_\ell}.$$

So the left side of the claim equals  $\sum_V \mu'^V \|L'_V F\|^2$ , the sum over selections.

Fix a selection  $V$  and pass to the coarser product space whose coordinates are, for  $(\ell, j) \in V$ , the head  $(\delta_{\ell,i})_{i < j}$  and the tail block  $B_{\ell,j}$ , and, for the other  $\ell$ , the whole  $X_\ell$ . There  $L'_V$  is the operator  $L_{V'}$  of Section 2 for the set  $V'$  of tail blocks. Each event  $E$  is a disjoint union of cylinder events on the coarse coordinates with the same coarse support: enumerate the values of the unfixed digits of each coarse coordinate that  $E$  involves. Splitting every event in this way does not change  $F$ , and exactly one piece of  $E$  holds at  $x$  if  $E$  does. The trace of the coarse support of  $E$  on  $V'$  is the set of the  $B_{\ell,j}$  with  $(\ell, j) \in V$  and  $j < v_\ell(E)$ . Identify  $B_{\ell,j}$  with the pair  $(\ell, j)$  and put  $\hat{E} = \{(\ell, j) : i_0(\ell) \leq j < v_\ell(E)\}$ ; then this trace is  $\hat{E} \cap V$ . Since  $N$  depends only on traces, Lemma 3.1 gives

$$\|L'_V F\|^2 \leq \mathbb{E}_x [N_{\hat{\mathcal{H}}(x)}(V)^2], \quad \hat{\mathcal{H}}(x) = \{\hat{E} : E \in \mathcal{E}, x \in E\}.$$

Sum over selections and add the nonnegative terms of all other sets  $V$  of pairs: the left side of the claim is at most  $\mathbb{E}_x [Q_{\mu'}(\hat{\mathcal{H}}(x))]$ . By Theorem 4.3 it remains to show that  $\prod_{(\ell,j) \in \hat{E}} (1 + \mu'_{\ell,j}) \leq 2$  for every event. Fix  $\ell$ , write  $\lambda = \lambda_\ell$ ,  $v = v_\ell(E)$ ,  $i_0 = i_0(\ell)$ , and assume  $v > i_0$ . Here  $1 + \mu'_{\ell,i_0} = \lambda^{i_0+1}$ , and  $1 + \mu'_{\ell,j} = 1 + \lambda^j(\lambda - 1)$  for  $j > i_0$  (if  $v \leq i_0$  the factor of  $\ell$  is 1). Every factor  $\lambda_\ell^{2v_\ell(E)}$  of the hypothesis is at least 1, so the hypothesis gives  $\lambda^{2v} \leq 2$  for each  $\ell$  separately, and  $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$  for  $j < v$ , which is equivalent to  $1 + \lambda^j(\lambda - 1) \leq \lambda^2$ . The factor of  $\ell$  is therefore at most  $\lambda^{i_0+1} \lambda^{2(v-1-i_0)} \leq \lambda^{2v}$ , and the product over  $\ell$  is at most  $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$ .  $\square$

**Corollary 7.2** (Modulus-weighted tail). *In Theorem 7.1 let the  $\ell \in \mathcal{P}$  be primes, let  $T > 1$ ,  $\mathcal{L} = \log T$ , and suppose that  $\prod_\ell \ell^{v_\ell(E)} \leq T$  for every*

event. For a set  $U$  of digits put  $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$ . Then for every  $\tau \geq 0$

$$\sum_{U: m_U > e^\tau} \|F^{=U}\|^2 \leq 2^{-\tau/(2\mathcal{L})}.$$

*Proof.* Take  $\lambda_\ell = 2^{\log \ell/(2\mathcal{L})}$ . Then  $\prod_\ell \lambda_\ell^{2v_\ell(E)} = 2^{\log \prod_\ell \ell^{v_\ell(E)}/\mathcal{L}} \leq 2$ , and the weight of  $U$  in Theorem 7.1 is  $2^{\log m_U/(2\mathcal{L})}$ ; apply Lemma 2.2(b).  $\square$

**Remark 7.3** (The factor 2 in the exponent). The proof gives more. If  $v > i_0$ , the factor of  $\ell$  is

$$\lambda^{i_0+1} \prod_{i_0 < j < v} (1 + \lambda^j(\lambda - 1)) \leq \exp\left(\lambda^{i_0+1} - 1 + \sum_{i_0 < j < v} \lambda^j(\lambda - 1)\right) = \exp(\lambda^v - 1).$$

So it suffices that  $\sum_\ell (\lambda_\ell^{v_\ell(E)} - 1) \leq \ln 2$  for every event. In Corollary 7.2 take  $\lambda_\ell = 2^{\log \ell/(\rho\mathcal{L})}$  and put  $x_\ell = v_\ell(E) \log \ell/\mathcal{L}$ , so that  $\sum_\ell x_\ell \leq 1$ . The function  $x \mapsto 2^{x/\rho} - 1$  is convex and vanishes at 0, so it is superadditive, and  $\sum_\ell (2^{x_\ell/\rho} - 1) \leq 2^{1/\rho} - 1$ . This is at most  $\ln 2$  when  $\rho \geq \rho_0 := 1/\log_2(1 + \ln 2) = 1.3163\dots$ , and then the tail in Corollary 7.2 is at most  $2^{-\tau/(\rho\mathcal{L})}$ .

## 8. APPLICATION: ERDŐS–STRAUS WITNESS MODULI

This section explains where Theorem 7.1 is used; the details are in [19], and nothing here is needed for the rest of the note. The Erdős–Straus conjecture asserts that  $4/n = 1/x + 1/y + 1/z$  has a solution in positive integers for every  $n \geq 2$ ; it is open. For  $M \equiv 3 \pmod{4}$  the classical multiplier identity gives a solution for every  $n$  with  $n \equiv -uv^{-1} \pmod{M}$  for some factorisation  $uvw = (M+1)/4$ . Let  $W(n)$  be the least such  $M$  ( $+\infty$  if there is none). In [19] it is shown, assuming Gallagher’s prime number theorem for the family of all characters and four estimates of Elsholtz and Tao on divisor sums, that  $W(p) \geq \exp(c(\log p)^{1/5}(\log \log p)^{-1/5})$  for infinitely many primes  $p$ . This is an  $\Omega$ -result about one particular family of explicit solutions, and says nothing about whether the conjecture holds.

The structure of the argument is as follows. Fix  $T$  and  $\mathcal{L} = \log T$ . After a *quarantine*  $n \equiv 1 \pmod{Q}$ , the condition  $W(n) > T$  holds as soon as  $n$  avoids a finite family of congruence events  $n \equiv b_i \pmod{r_i}$  with  $r_i \leq T$ , at most  $T^2$  of them. For  $n$  distributed by Haar measure on the unit group  $\widehat{\mathbb{Z}}^\times$ , conditioned on the quarantine, the residues  $X_\ell = n \bmod \ell^{f_\ell}$  at the primes involved are independent. Each  $X_\ell$  is uniform on the units modulo  $\ell^{f_\ell}$ , or on a fibre  $1 + \ell^a \mathbb{Z}$  modulo  $\ell^{f_\ell}$  if  $\ell$  is quarantined with  $a = a_\ell \geq 1$ . On these sets the uniform measure is the product of the uniform distributions of the free base- $\ell$  digits (digit 0 ranging over the nonzero residues in the unit case), so the digits are independent coordinates. Moreover an event  $n \equiv b \pmod{r}$  is a digit-prefix event with  $\prod_\ell \ell^{v_\ell(E)} = r \leq T$ . So the avoidance indicator  $F$  is in the setting of Corollary 7.2, and so is each conditioned function  $F^{(j)}$  (the indicator that no earlier event occurs, restricted to the cylinder of the  $j$ -th event): there the digit offsets  $i_0(\ell)$  are raised.

To find a *prime*  $p$  with  $W(p) > T$  one needs a minorant  $B \leq F$  that is a combination of congruence classes of small modulus, has mean comparable to  $\mathbb{E}F$ , and has small  $\ell^1$  norm. [19] uses the one-sided  $\ell^2$  sandwich of Bazzi [2] in the form of Razborov [17]:

$$B = 1 - \sum_i \mathbf{1}_{E_i} \left( 1 - \sum_{j < i} \mathbf{1}_{E_j} u_j \right)^2 \leq F,$$

$$\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}[(F^{(j)} - u_j)^2],$$

for any functions  $u_j$  of the coordinates not fixed by  $E_j$ . The natural choice of  $u_j$  is the truncation of the Efron–Stein decomposition of  $F^{(j)}$  to digit sets  $U$  with  $m_U \leq e^\tau$ . By (1) it is a combination of classes of modulus at most  $e^\tau$  (so  $\mathbf{1}_{E_j} u_j$  has modulus at most  $T e^\tau$ , and every class in the expansion of  $B$  has modulus at most  $T^3 e^{2\tau}$ ), and by Corollary 7.2 its error is at most  $2^{-\tau/(2\mathcal{L})}$ . The sandwich needs the error to be about  $e^{-O(S)} m^{-2}$ , where  $S$  is the total mass of the events. So  $\tau \ll \mathcal{L}(S + \mathcal{L})$  suffices; in [19]  $S \ll \mathcal{L}^4$  (using the Elsholtz–Tao input), and the cell moduli  $d_i$  of  $B$  satisfy  $\log d_i \leq 3\mathcal{L} + 2\tau \ll \mathcal{L}^5$ . The quarantine costs  $\log Q \ll \mathcal{L}^5 \log \mathcal{L}$ , and a linear transfer theorem based on Gallagher’s theorem then produces a prime  $p$  with  $\log p \ll \mathcal{L}^5 \log \mathcal{L}$ .

If instead one encodes the residues in binary and bounds the Fourier tail of  $F^{(j)}$  by the switching-lemma route of [14], the junta moduli pick up factors for the width of the events (the number of primes involved) and for the bit length of the residues, since a junta on  $d$  bits may read the top bit of each residue. Theorem 7.1 avoids both factors. The exponent  $1/5$  needs two further ingredients of [19] (a graded quarantine and a moment bound), and we do not attribute it to Theorem 7.1 alone. Two features of the energy bound matter here: the events are congruences modulo integers up to  $T$ , so the alphabets are huge and the measure is far from the Boolean setting, and the cost of a junta must be measured by its modulus rather than by the number of primes it involves.

## 9. COMPUTATIONAL CHECKS

The proofs above are complete. The script `scripts/energy_note_check.py` is a sanity check of the statements on small cases. It uses exact rational arithmetic throughout; where a statement involves an irrational number, the comparison is rewritten as an exact rational one (for instance  $W^{>t} \leq c 2^{-(t+1)/k}$  as  $(W^{>t})^k 2^{t+1} \leq c^k$ ). With seed 1 it checks:

- (A) Theorem 1.1, Lemma 2.2(a) and Lemma 3.1 (for every  $V$ ) on 150 random systems on  $\prod_v [q_v]$  ( $n \leq 4$  coordinates,  $q_v \leq 5$ ), with random rational product measures and random rational weights in  $[1, 2]$ ;
- (B) the identity of Lemma 4.1, the two formulas for  $N_{\mathcal{H}}$ , and the matching bounds of Lemma 4.2 and Theorem 4.3 (including the equality case) on 300 random weighted hypergraphs on at most 5 vertices;

- (C) Corollary 1.3 for *every* Boolean function on 4 bits under the uniform measure, with  $k$  its minimal DNF width: the tail bound for all  $t$  (exactly), the bounds  $1 + 4p$  and  $(4k/\ln 2)p$  at the endpoint  $\lambda = 2^{1/k}$  (exactly for  $k = 1$ , in floating point with slack  $10^{-12}$  otherwise), and  $G_F(\lambda) \leq 1$  at a rational  $\lambda \leq 2^{1/k}$ ; the largest value of  $W^{>t}[g] 2^{(t+1)/k}/(4p(2-p))$  is 0.680;
- (D) the same statements on 150 random DNFs under biased product measures on  $\{\pm 1\}^n$  and  $q$ -ary DNFs with set-valued literals;
- (E) Theorem 7.1 on 150 random digit-prefix systems in bases 2 and 3, with random offsets  $i_0(\ell) \in \{0, 1\}$  and random digit distributions;
- (F) the level polynomial of Proposition 6.1 against brute force, Corollary 1.2 along that family, the formula of Example 6.2, and the sign in both instances of Example 6.3 (exact on  $[8]^2$ , 50-digit on  $[5]^3$ ). The values  $(\max_s \text{En}(F; jk - 1))^{1/j}$  of Example 6.2 (0.299, 0.311, 0.319 for  $j = 10, 20, 40$ ) are printed, not checked.

The run takes about six minutes on one core; its output is in the file `data/energy_note/check_seed1.txt`.

#### REFERENCES

- [1] K. Amano, Tight bounds on the average sensitivity of  $k$ -CNF, *Theory Comput.* 7 (2011), 45–48.
- [2] L. M. J. Bazzi, Polylogarithmic independence can fool DNF formulas, *SIAM J. Comput.* 38 (2009), 2220–2272 (FOCS 2007).
- [3] P. Beame, A switching lemma primer, Technical Report UW-CSE-95-07-01, University of Washington, 1994.
- [4] E. Blais, R. O’Donnell and K. Wimmer, Polynomial regression under arbitrary product distributions, *Mach. Learn.* 80 (2010), 273–294 (COLT 2008).
- [5] R. B. Boppana, The average sensitivity of bounded-depth circuits, *Inform. Process. Lett.* 63 (1997), 257–261.
- [6] E. Chattopadhyay, P. Hatami, K. Hosseini and S. Lovett, Pseudorandom generators from the second Fourier level and applications to  $\text{AC}^0$  with parity gates, in: *Proc. 10th ITCS* (2019).
- [7] B. Efron and C. Stein, The jackknife estimate of variance, *Ann. Statist.* 9 (1981), 586–596.
- [8] M. L. Furst, J. C. Jackson and S. W. Smith, Improved learning of  $\text{AC}^0$  functions, in: *Proc. 4th COLT* (1991), 317–325.
- [9] J. Håstad, Almost optimal lower bounds for small depth circuits, in: *Proc. 18th ACM STOC* (1986), 6–20.
- [10] J. Håstad, A slight sharpening of LMN, *J. Comput. System Sci.* 63 (2001), 498–508.
- [11] J. Håstad, On the correlation of parity and small-depth circuits, *SIAM J. Comput.* 43 (2014), 1699–1708.
- [12] W. Hoeffding, A class of statistics with asymptotically normal distribution, *Ann. Math. Statist.* 19 (1948), 293–325.
- [13] V. Lecomte and L.-Y. Tan, Sharper bounds on the Fourier concentration of DNFs, in: *Proc. 62nd IEEE FOCS* (2021), arXiv:2109.04525.
- [14] N. Linial, Y. Mansour and N. Nisan, Constant depth circuits, Fourier transform, and learnability, *J. ACM* 40 (1993), 607–620.
- [15] Y. Mansour, An  $O(n^{\log \log n})$  learning algorithm for DNF under the uniform distribution, *J. Comput. System Sci.* 50 (1995), 543–550.

- [16] R. O'Donnell, Analysis of Boolean Functions, Cambridge University Press, 2014.
- [17] A. Razborov, A simple proof of Bazzi's theorem, ACM Trans. Comput. Theory 1 (2009), no. 1, Art. 3.
- [18] B. Rossman, An entropy proof of the switching lemma and tight bounds on the decision-tree size of  $AC^0$ , manuscript, 2017.
- [19] Anonymous, Large Erdős–Straus witnesses II: the multiplier modulus is infinitely often  $\exp((\log p)^{1/5-o(1)})$ , preprint, 2026.
- [20] A. Tal, Tight bounds on the Fourier spectrum of  $AC^0$ , in: Proc. 32nd CCC (2017), LIPIcs 79, Art. 15; ECCC TR14-174.